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Do environmental scores become multinational corporations' strategic “greenwashing” tool for window-dressing carbon reduction? A cross-cultural analysis

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Abstract

Owing to the urgency of the climate change crisis and the emerging greenwashing issue that companies face, greenwashing in companies' carbon reduction is exceptionally important in the progress of global carbon reduction and the achievement of the Paris Agreement goals. Therefore, this study focused on multinational corporations (MNCs) listed on the 2021 Fortune Global 500 list (2001–2020) and used institutional theory to theoretically discuss and empirically test the relationship between MNCs' environmental scores and carbon emissions/intensity, as well as the moderating effect of national culture. Our results demonstrate that the environmental scores of MNCs positively affect their carbon emissions and intensity, meaning that managers tend to use high scores to hide their poor carbon reduction, showing a typical “talk” more and “act” less (i.e., greenwashing) behavior. Meanwhile, the moderating effects of individualism, masculinity, and uncertainty avoidance were negative. These revealed that the balance of MNCs' “talk” and “act” can be influenced by the domestic national cultural characteristics. Further analysis showed that the Paris Agreement did not prevent MNCs' greenwashing in carbon reduction. Such greenwashing behavior has become even worse in societies with power distance characteristics after the Paris Agreement. To the best of our knowledge, this is the first study to investigate companies' greenwashing in terms of carbon reduction. The goal is to enhance our understanding of why and how greenwashing occurs based on the institutional theory perspective, extend the application of institutional theory, enrich the literature on companies' greenwashing and national culture, and offer insight into interdisciplinary research between national culture and green transition management. Finally, suggestions for policymakers on how to prevent MNCs' greenwashing in carbon reduction, considering national cultural influence, were provided.

Abbreviations: CSR, corporate social responsibility; MNCs, multinational corporations.

Xuhui Peng and Jingduan Li did same contributions in this study.

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KEYWORDS

carbon reduction, environmental score, greenwashing, multinational corporations, national culture

1 | INTRODUCTION

Over the past few decades, the consequences of climate change have attracted significant attention in academia and other practical spheres. According to the Intergovernmental Panel on Climate Change (IPCC, 2021), climate change is caused by increased carbon emissions from human activities. Reportedly, climate change has caused a 1.59°C increase in surface temperature, rising sea levels, and decreasing sea ice concentration in the Arctic and Antarctic regions (IPCC, 2021), thus leading to extreme weather,¹ frequent natural disasters,² ecosystem damage, and an unsustainable food crop supply (Rigaud et al., 2018). The World Bank suggests that without further implementation, these crises will threaten the world's agricultural areas and generate more than 143 million climate change refugees by 2050 (Rigaud et al., 2018). Thus, carbon emissions should be carefully considered, particularly in the private sector.

The private sector has received academic attention owing to its environmental performance as a major contributor to carbon emissions. The existing literature has shown that companies' environmental performance reflects environmental responsibility-related strategies and practices (Hussain et al., 2018), which can explain their responses to climate change (Galbreath, 2010). With increasing climate change concerns from stakeholders (He et al., 2021), companies must focus on their environmental responsibility for climate change and ensure that related information disclosures comply with governmental regulations (Zhou et al., 2022), social norms (Marano & Kostova, 2016; Matten & Moon, 2008), and stakeholder demands (He et al., 2021). In turn, advantageous performance can help companies strengthen their stakeholder connections (Grisard et al., 2020; Zhou et al., 2022), avoid related regulatory penalties (Li & Lu, 2020; Zhou et al., 2022), and enhance corporate image (Garriga & Melé, 2004), which enhances investors' valuation (Awaysheh et al., 2020) and precipitates improved financial performance (Choi et al., 2021).

Previous studies have been skeptical of corporate environmental performance scores. Hussain et al. (2018) argue that companies' environmental responsibility information can be found in corporate social responsibility (CSR)-related reports. These public disclosures became the data source for environmental, social, and governance (ESG) rating providers (e.g., Thomson Reuters) to evaluate their environmental performance scores for the companies (Young-Ferris & Roberts, 2021). Typically, more comprehensive environmental responsibility disclosures could lead to a better evaluated environmental performance score, which becomes a channel for stakeholders to understand companies' participation in environmental responsibility (Fernandez-Feijoo et al., 2014) and help investors make investment decisions (Young-Ferris & Roberts, 2021). However, the existing

regulatory system does not have mandatory independent audit requirements for companies' environmental statements (Ramus & Montiel, 2005). This may increase the uncertainty of corporate environmental disclosure (Torelli et al., 2020), and the credibility of the related data is difficult to ensure (Ramus & Montiel, 2005), which directly affects the ESG rating providers' evaluation of companies. Consequently, stakeholders' perceptions of the companies can be misled by the suspectable environmental performance (Torelli et al., 2020), and such corporate behavior has been termed "greenwashing" in previous studies (Ramus & Montiel, 2005; Seele & Gatti, 2017; Siano et al., 2017; Torelli et al., 2020).

Thus, academic interest in greenwashing has increased. Torelli et al. (2020) argued that companies' greenwashing mainly covers corporate disclosure at the corporate, strategic, dark, and product levels, and the key characteristic is achieving specific goals by communicating with stakeholders. A discrepancy among companies' declarations and their actual practice is a typically greenwashing behavior, that is, "talk" more and "act" less (Siano et al., 2017). This includes a series of intentional impression management practices to earn benefits for companies from the created green image without actual action. Siano et al. (2017) suggested that such greenwashing behavior is divided mainly into decoupling and attention-deflection types. Specifically, companies may adopt a set of empty environmental claims or policies to decouple greenwashing (Ramus & Montiel, 2005) and cause the sin of fibbing. In attention-deflection greenwashing, companies normally choose to disclose selected information (Lyon & Montgomery, 2015) or uncertain information, which includes disclosing positive information and concealing negative one (Torelli et al., 2020) or exaggerating their environmental practices (Lyon & Montgomery, 2015). In doing so, companies' environmental faults can be hidden, and competitiveness can be promoted (Balluchi et al., 2020). Clarkson et al. (2008) argued that companies have widely adopted this information disclosure as a "signal" to respond to stakeholders' concerns. Decoupling or attention-deflection greenwashing, the core feature of greenwashing, misleads stakeholders through company information disclosure.

Studies have investigated the motivations for, and consequences of, companies' greenwashing. Seele and Gatti (2017) stated that companies could strategically mislead stakeholders (i.e., greenwashing) to meet their legitimacy needs because the claims potentially benefit the company (the same as an actual corporate practice). More importantly, it is easier to use at a low cost (Walker & Wan, 2012), especially when a company faces tremendous public pressure (De Villiers & Van Staden, 2006; Yu et al., 2020). What is worse, the corporate profit-driven nature (De Vries et al., 2015; Du, 2015) and managers' personal interests (Siano et al., 2017) encourage misleading behavior. For example, the current remuneration system tempts managers to focus more on short-term economic interests and less on environmental responsibilities (Bai & Elyasiani, 2013). Companies' desire for profits

¹Fidler et al. (2022).

²Further information can be found here: <https://www.climatechange.environment.nsw.gov.au/storms-and-floods>.

drives them to adopt environmental claims to earn stakeholder trust (Torelli et al., 2020) and enhance their financial returns from the capital market (Awaysheh et al., 2020). Previous studies investigated the impact of companies' environment-related greenwashing on stakeholder perceptions (Torelli et al., 2020), corporate legitimacy (Berrone et al., 2017; Seele & Gatti, 2017), brand trust (Guo et al., 2017), and profitability (Du, 2015). Companies' irresponsible environmental claims have caused a series of potential ethical harms (Nyilasy et al., 2014) that should be questioned (Rahman et al., 2015). However, studies on corporate greenwashing in terms of carbon emissions are yet to be conducted.

Furthermore, the importance of national culture in companies' environmental responsibility has emerged recently (Luo & Tang, 2022; Peng et al., 2022, 2023; Peng & Zhang, 2022; Ullah et al., 2022) and needs to be considered in greenwashing research. Hofstede et al. (2010) posited that individual moral cognition and social values are determined by national culture worldwide, which directly leads to diverse political, economic, and social norms between regions and forms different institutional pressures. Companies' leadership styles (Ardichvili & Kuchinke, 2002), managers' business ethics (Okpara, 2014), and environmental concerns can differ in diverse national cultural contexts (Luo & Tang, 2016). Therefore, national culture is the most important normative isomorphic institutional factor in corporate environmental responsibility (Peng & Zhang, 2022). Companies' environmental practices are the outcome of strategic decision-making (Peng et al., 2022; Peng & Zhang, 2022), which is determined by the company's high governance body (e.g., the board of directors) (Rao & Tilt, 2016). Under these circumstances, the normative isomorphic force of national culture may result in multiple institutional factors (e.g., social norms and regulations) that constrain corporate environmental responsibility, while the environmental concerns of high governance bodies are also influenced. This has made national culture one of the most important institutional factors in corporate environmental responsibility (Luo & Tang, 2022; Peng et al., 2023; Peng & Zhang, 2022). The existing literature has proven the role of national culture in corporate carbon performance (Luo & Tang, 2022) and the moderating effect of national culture on board composition and environmental performance (Peng et al., 2023; Peng & Zhang, 2022). Regarding companies' greenwashing, the influence of national culture on individual moral cognition is likely caused by diverse managers' strategic greenwashing behaviors. However, these findings require further investigation.

Considering the current climate change issue and the above arguments and evidence, we argue that under the great public climate change pressure, companies may adopt better environmental claims to mislead stakeholders. However, their actual effects on carbon reduction remain unknown. Such greenwashing not only causes ethical harm but also creates a barrier to the progress of the global climate change response. However, current studies mainly focus on companies adopting CSR disclosures to mislead stakeholders and other greenwashing behaviors in environmental responsibility (Testa et al., 2018; Torelli et al., 2020) or business scandals (Siano et al., 2017), while academic attention on companies' greenwashing in

carbon reduction is lagging. Clarkson et al. (2008) suggested that future research should focus on inquiries beyond disclosure. Therefore, following Choi and Luo (2020) and Luo and Tang (2022), corporate carbon reductions should consider both absolute emissions and intensity reductions. In doing so, we focus on multinational corporations (MNCs) listed on the 2021 Fortune Global 500 list and investigate the relationship between corporate environmental scores and carbon emissions (intensity) to determine whether large companies with higher environmental scores (i.e., "talk" more) actually participate in carbon reduction (i.e., "act" more) or not (i.e., "act" less). From an institutional theory perspective, we expect to explain theoretically and empirically test how and why environmental scores can become strategic tools for MNCs to mislead stakeholders as a new form of greenwashing to address their failure in carbon reduction. Considering the role of national culture in managers' moral cognition and corporate environmental responsibility, this study also investigated the moderating effect of national culture on corporate environmental scores and carbon emissions. The reason for doing so is that we expect to reveal how diverse national cultural contexts could affect managers' moral cognition, which could encourage or constrain their greenwashing decisions.

Finally, this study extends the application of institutional theory and the national culture dimension to business ethics and management. Moreover, suggestions for policymakers to restrain MNCs' strategic greenwashing behavior by considering national culture are based on our results. The remainder of this paper is structured as follows. Section 2 presents the literature review and hypotheses development, Section 3 presents the methodology of this study, Section 4 presents our empirical results, and Section 5 presents the discussion.

2 | LITERATURE REVIEW AND HYPOTHESES

The institutional theory explains why institutional pressure affects corporate practices. DiMaggio and Powell (2000) stated that institutional pressures are generated by different institutional factors that determine legitimacy standards for companies, including government regulation (Christensen et al., 2022; Li & Lu, 2020), the scrutiny of trading associations (Berrone et al., 2017), social value (Guthrie et al., 2004; Kostova & Roth, 2002), and national culture (Peng & Zhang, 2022). Under this pressure, companies must adjust or restructure their business strategies to ensure that their practices comply with regulatory requirements (Li & Lu, 2020), respond to stakeholder demands (Berrone et al., 2017; Torelli et al., 2020), and are consistent with social norms (Marano & Kostova, 2016). Meanwhile, national culture is an essential institutional factor affecting individual moral reasoning, politics, and leadership ethics concerns (Hofstede et al., 2010), thus precipitating different management styles (Morris & Pavett, 1992) and managers' environmental concerns (Luo & Tang, 2016). Therefore, companies' decision-making regarding environmental responsibility can differ in diverse national and cultural contexts (Peng et al., 2023; Peng & Zhang, 2022).

Considering these points, MNCs' strategic greenwashing behavior of MNCs in response to climate change and the influence of national culture can be explained from an institutional theory perspective.

2.1 | MNCs' greenwashing

Increased public awareness exacerbates MNCs' institutional pressure regarding climate change. The existing literature posits that corporate financial performance, resources, and reputation can be determined by legitimacy (DiMaggio & Powell, 2000; Walker & Wan, 2012), which is directly linked to corporate development and survival (DiMaggio & Powell, 2000), depending on whether the company can adopt appropriate corporate practices to relieve the institutional pressures generated by diverse stakeholders (Kostova & Roth, 2002; Marano & Kostova, 2016). In these circumstances, the perception of a company's stakeholders has become key to maintaining legitimacy (Torelli et al., 2020), especially in environment-related issues. Regarding emerging public climate change concerns, the call for companies to reduce carbon emissions has become a priority since the Kyoto Protocol (He et al., 2021). This has prompted governments and nongovernmental organizations to impose additional institutional pressure on companies (Beddewela & Fairbrass, 2016; He et al., 2021), including climate change-related government regulations (Li & Lu, 2020), the Emissions Trading Scheme (Ferguson et al., 2016), and additional requests from the Securities Regulatory Commission (Yang et al., 2021; Zhou et al., 2018) and other stakeholders (Luo & Tang, 2022; Wang et al., 2021; Zhou et al., 2022). Therefore, companies are required to adjust their strategies and participate in appropriate practices to relieve pressure (Peng et al., 2023; She & Michelon, 2019), including increasing investments in reducing carbon emissions.

However, this investment may be limited by a desire for profit. Berrone et al. (2017) stated that corporate environmental endeavors impose high costs on companies because such corporate practices require them to invest more time and capital (Berrone et al., 2013), which increases companies' financial pressure in the short term (López et al., 2007), especially in carbon emission reduction. For example, the cost of CO₂ capture per metric ton of work consumption is extremely high (average between 20.5 and 107.9 US dollars) (Zang et al., 2021). However, the outcome of these investments typically takes a long time to influence the related performance (Berrone et al., 2013), and the return is unlikely to meet expectations (Makni et al., 2009). Similarly, the companies' responses to the CDP questionnaire also demonstrate that the companies' carbon emission reduction-related investments (e.g., energy efficiency and low-carbon energy) generally need long-term returns, some of them even longer than their lifetime. Moreover, the current remuneration system for managers is linked to corporate financial performance, which may reduce managers' environmental concerns and allow them to focus more on short-term financial interests (Bai & Elyasiani, 2013). Thus, investment in environmental responsibility is unlikely to meet companies' needs.

Companies may adopt strategic greenwashing behavior for legitimacy and profits. Budinsky and Bryant (2013) posited that the current

system might result in greenwashing issues because mandatory regulations pertaining to independent auditing for voluntary disclosure (Ramus & Montiel, 2005) and economic punishment for companies that cannot achieve their environmental commitment are lacking (Berrone et al., 2017). This may provide an opportunity for companies to conceal negative information and disclose the positives (Torelli et al., 2020), even proposing environmental claims to mislead stakeholders (Ramus & Montiel, 2005); thus, companies may adopt strategic information disclosure to exaggerate environmental performance without actual actions (Lyon & Montgomery, 2015) and mislead stakeholders for legitimacy maintenance (Siano et al., 2017). Over 86% of large MNCs' sustainability practices are imbalanced in their commitment goals and are less transparent about their negative impacts (KPMG, 2020). This intentional misleadership results in the deceptive manipulation of greenwashing and causes ethical harm (Siano et al., 2017).

Recent studies have provided evidence of companies' greenwashing behavior. Siano et al. (2017) investigated the Volkswagen scandal and demonstrated that the company intentionally adopted exaggerated information to mislead stakeholders about the failure of its decarbonization commitment. Torelli et al. (2020) analyzed the impact of greenwashing on stakeholders' perceptions and found that companies could affect their perceptions by implementing strategic communication (greenwashing) for legitimacy purposes. Walker and Wan (2012) demonstrated how greenwashing affects corporate financial performance, and De Vries et al. (2015) confirmed that profitability is among the most important motivations for corporate greenwashing. Seele and Gatti (2017) confirmed the effect of greenwashing as a strategy for corporate legitimacy.

Considering these findings, we follow the arguments of Ramus and Montiel (2005), Berrone et al. (2017), Siano et al. (2017), and Torelli et al. (2020), assuming that increasing public concern regarding climate change has imposed great institutional pressure on MNCs, requiring them to participate in carbon reduction-related practices to relieve pressure. However, the costs of such practices (e.g., energy efficiency and low-carbon energy) are normally higher than the returns, which is contrary to MNCs' desire for profits. Under these circumstances, a lack of voluntary disclosure auditing and economic punishment for failing to meet environmental commitments may allow MNCs to adopt exaggerated environmental statements, such as greenwashing. This may become a strategic tool for companies to mislead stakeholders when addressing the failure of carbon reduction for profit and legitimacy purposes. Consequently, MNCs with poor carbon reduction (i.e., high absolute carbon emissions and intensity) are likely to exhibit higher environmental scores. Thus, we propose the following hypotheses:

Hypothesis 1a. There is a positive relationship between MNCs' environmental scores and absolute carbon emissions.

Hypothesis 1b. There is a positive relationship between MNCs' environmental scores and carbon intensity.

2.2 | National culture

Corporate practices may be determined by the national culture. Hofstede et al. (2010) introduced the concept of national cultural dimensions (power distance, individualism, masculinity, uncertainty avoidance, long-term orientation, and indulgence) to differentiate general individual moral principles and social norms across regions. In particular, culture affects individual concerns about values (Hofstede et al., 2010), managers' attitudes toward rules and self-interest (Okpara, 2014), and subordinates' submission and loyalty (Patel et al., 2002). These differences result in different sustainable consumption demands (Manta et al., 2022), management styles (Ardichvili & Kuchinke, 2002; Manta et al., 2023; Westerman et al., 2007), corporate strategies focused on long- or short-term interests (Peng & Zhang, 2022), and definitions of business success (Luo & Tang, 2022). The stakeholders' demands also differ according to cultural backgrounds (Ringov & Zollo, 2007). These generate significant institutional pressure and form diverse legitimacy standards for companies, which require them to participate in appropriate practices to meet these standards, especially for CSR-related issues (Marano & Kostova, 2016). Thus, companies' practices are heterogeneous under the influence of national culture, which further determines disclosure quality (Vitolla et al., 2019).

Ethical managerial decision-making can be influenced by national culture at both the macrolevels and microlevels (Westerman et al., 2007). Kymlicka (2020) argued that culture creates institutional pressure on society members to restrain their behavior and comply with social norms to avoid the risk of social membership loss. This resulting in differences in managers' behavioral intentions, attitudes, and acceptance of business practices (e.g., bribery condonement) in different cultural contexts (Westerman et al., 2007). According to Westerman et al. (2007), managerial ethical responses normally follow the social identity referent, resulting in different ethical business decision-making between diverse national cultural backgrounds. Moreover, previous studies have argued that companies' environmental concern (Luo & Tang, 2016), the board's attention to stakeholders' environmental demands (Peng & Zhang, 2022), and the ability to detect corporate water responsibility issues differ in different cultural environments (Peng et al., 2023).

Therefore, companies' greenwashing behavior in climate change is the outcome of ethical corporate decision-making, which is likely influenced by the role of national culture in managers' moral principles and social norms.

2.2.1 | Power distance

A centralized management style can be generated by increasing the power distance. Hofstede et al. (2010) posited that power distance describes the tolerance levels of society members in an unequal power distribution. Specifically, less powerful members are unlikely to question authority (Weaver, 2001), and subordinates tend to present a submission principle with increasing power distance (Patel

et al., 2002). This may discourage subordinates from participating in formal corporate governance (Li & Harrison, 2008) and from blindly following orders (Morris & Pavett, 1992), even against ethical principles (Okpara, 2014). Consequently, authorities in such societies are highly likely to ignore others' suggestions and form a centralized management style, and decision-making power may only be handled by them (Hofstede et al., 2010).

Corporate ethical decision-making may worsen because of the autocratic power of high-power distance. As discussed above, under the influence of high-power distance, authorities have been provided with extremely high power in management and decision-making, which may have formed an autocratic power for top management (Rowley & Bae, 2002), thus reducing the objectivity of decision-making (Peng & Zhang, 2022) and affecting businesses' ethical behavior intentions (Westerman et al., 2007). The existing literature has also argued that, compared to a low-power distance society, companies from a high-power distance society are more likely to exhibit lower board management supervision (Cui et al., 2020), greater interest in short-term financial gains (Peng & Zhang, 2022), and less concern regarding stakeholders' environmental demands (Peng et al., 2023). Therefore, corporate ethical decision-making may be weakened by high-power distance, especially in environment-related issues. Previous evidence has also demonstrated that high-power distance negatively affects corporate social performance (Ringov & Zollo, 2007), CSR engagement (Peng et al., 2012), and environmental innovation (Ullah et al., 2022), and companies from high-power distance contexts normally present higher carbon emission intensity (Luo & Tang, 2022).

Considering the above, we follow the arguments by Westerman et al. (2007), Hofstede et al. (2010), Okpara (2014), and Peng and Zhang (2022) and assume that the influence of power distance on individuals can strengthen the autocratic power of top management and generate a submission principle for subordinates. This may amplify managers' desire for profit without doubt from others and encourage them to adopt an exaggerated environmental statement to mislead stakeholders rather than actual participation in carbon reduction. Thus, we propose the following hypotheses:

Hypothesis 2a. Power distance positively moderates the relationship between environmental scores and absolute carbon emissions of MNCs.

Hypothesis 2b. Power distance positively moderates the relationship between MNCs' environmental scores and carbon intensity.

2.2.2 | Individualism

Public pressure for climate change concerns that companies undertake may increase in societies with high levels of individualism. Hofstede et al. (2010) posited that individualism can explain individual differences in interest recognition (oneself or other members of

society). Specifically, with an increase in individualism, members of society tend to be more concerned about their own interests and those of their family members (Hofstede et al., 2010). Meanwhile, individuals from high-individualism societies normally take more initiative in proposing demands (Husted, 2005), and a social mechanism is also exhibited in an open society with competition and innovation (Jaggi & Low, 2000). This may lead to societies with high individualism that tend to emphasize individual responsibility and self-empowerment (Disli et al., 2016), which may encourage stakeholders to propose their demands, and environmental issues will be discussed publicly. Furthermore, this may generate significant public pressure on authorities to enhance companies' environmental accountability (Husted, 2005).

Increasing public pressure may enhance ethical corporate decision-making in highly individualistic societies. According to Husted (2005) and Disli et al. (2016), societies with high individualism emphasize individual responsibility and environmental accountability. These individuals are more concerned with the agent's responsibility and ethical policies (Scholtens & Dam, 2007); therefore, companies from such societies are likely to participate in environmental innovation and other forms of corporate social involvement (Vachon, 2010). Conversely, societies with low individualism create a collaborative culture where companies promote internal communication (Mitchell et al., 2000) and encourage managers to avoid conflicts with insiders (Newbury & Yakova, 2006). However, these collectivists invariably favor outsiders' expenses, which may reduce their insensitivity to ethical issues and cause irresponsibility issues (e.g., corruption; Davis & Ruhe, 2003; Husted, 2000; Peng & Lin, 2009); thus, individualism may promote ethical corporate decision-making. Further, existing evidence indirectly supports this argument: Scholtens and Dam (2007) indicated that individualism positively impacts business ethics; Vachon (2010) reported that individualism positively affects companies' environmental practices; and Disli et al. (2016) found a negative relationship between individualism and carbon emissions.

Considering the above, we follow Husted (2005), Scholtens and Dam (2007), and Disli et al. (2016) and assume that individualism can enhance individuals' responsibility and companies' environmental accountability, which may generate great public pressure on companies regarding climate change and strengthen corporate ethical decision-making. Under these circumstances, managers may exhibit higher ethical standards and participate in carbon reduction rather than strategic greenwashing. Thus, we propose the following hypotheses:

Hypothesis 3a. Individualism negatively moderates the relationship between environmental scores of MNCs and absolute carbon emissions.

Hypothesis 3b. Individualism negatively moderates the relationship between MNCs' environmental scores and carbon intensity.

2.2.3 | Masculinity

Managers' desires for business success may be maximized in societies with high masculinity. Hofstede et al. (2010) posited that masculinity explains how society's definition of success tends to achieve heroism or harmoniousness. Compared to low-masculinity societies, individuals from high-masculinity societies emphasize ambition and competition (Hofstede et al., 2010). This creates a social norm that can influence managers to be more concerned about their career development (Peng et al., 2012) and encourage them to achieve business success (Santema et al., 2005). Moreover, they enhance managers' abilities in problem-solving and self-determination (Chinta & Capar, 2007), and the board tends to support management behaviors in a performance orientation (Li & Harrison, 2008). Therefore, managers' desire for success is likely to be strengthened in societies with high masculinity.

Corporate ethical decision-making may be improved by managers' desire for success in high-masculinity societies. This is because stakeholders' concerns about corporate irresponsibility are emerging (Siano et al., 2017), directly affecting their perception of business scandals (e.g., greenwashing) and threatening corporate legitimacy (Torelli et al., 2020). Meanwhile, under the influence of high masculinity, managers exhibit a passion for ambition and competitiveness in seeking success (Ullah et al., 2022), which causes them to consider fewer unethical practices, show higher moral values (Okpara, 2014), and focus more on stakeholders' needs for long-term business success. Okpara (2014) found that masculinity is positively related to managers' moral values, Ullah et al. (2022) verified that masculinity positively affects corporate environmental innovation, and Ho and Taylor (2007) reported that companies' triple bottom-line disclosure in a high-masculinity country (e.g., Japan) is better than in a lower-masculinity country (e.g., the United States).

Considering the aforementioned points, we follow Hofstede et al. (2010), Okpara (2014), Torelli et al. (2020), and Ullah et al. (2022) in assuming that managers from a high-masculinity cultural environment normally show stronger success than others, which causes them to achieve long-term business success. This may make them more concerned about stakeholders' perceptions of business scandals (e.g., talking without action) and less concerned about unethical decision-making. Therefore, under increased climate change pressure from stakeholders, managers from high-masculinity societies are more likely to participate in carbon reduction rather than adopt strategically exaggerated environmental statements. Thus, we propose the following hypotheses:

Hypothesis 4a. Masculinity negatively moderates the relationship between MNCs' environmental scores and absolute carbon emissions.

Hypothesis 4b. Masculinity negatively moderates the relationship between MNCs' environmental scores and carbon intensity.

2.2.4 | Uncertainty avoidance

Companies' operations are more structured with increased uncertainty avoidance, which may form a detailed management style. Hofstede et al. (2010) suggested that uncertainty avoidance reflects the extent to which an unknown situation or event in the future can result in anxiety or emotions among society members. Specifically, individuals may become more anxious and worried about their future due to increased uncertainty avoidance (Hofstede et al., 2010). Members of high-uncertainty avoidance societies tend to establish ritualistic and customary behavior standards for anxiety relief (Hofstede et al., 2010), provide more detailed information to employees (Li & Harrison, 2008), and adopt structured processing in management (Joiner, 2001) with direct control (Griffin et al., 2017) to stabilize them. Therefore, previous studies have suggested that managers from high-uncertainty avoidance societies normally prefer to follow rules without discretion (Griffin et al., 2017), and management styles tend to be detailed and structured (Li & Harrison, 2008).

The influence of high-uncertainty avoidance on top management may promote ethical decision-making. As previously discussed, a detailed management style (Li & Harrison, 2008) and manager rule-following (Griffin et al., 2017) are preferred in societies with high-uncertainty avoidance, potentially reducing the possibility of managers opposing the code of ethics. Meanwhile, previous studies have also argued that in such societies, managers exhibit cautious calculations of risk (Hofstede et al., 2010) to control their futures (Offermann & Hellmann, 1997) and may, thus, exhibit higher moral values (Okpara, 2014) and consider the risk of a negative influence from business scandals on stakeholders, which strengthens corporate ethical decision-making. This argument is indirectly supported by existing evidence: Okpara (2014) reported that uncertainty avoidance is positively related to managers' moral values and personal attributes, and Luo and Tang (2016) indicated that companies from high-uncertainty avoidance societies exhibit higher carbon disclosure propensity than those with low-uncertainty avoidance.

Therefore, we follow the arguments of Li and Harrison (2008), Hofstede et al. (2010), and Griffin et al. (2017) to assume that under the influence of high-uncertainty avoidance, managers' rule-following characteristics and their cautious calculation of risk for future security may enhance their awareness of business scandal risk. This may reduce the possibility of managers adopting strategic greenwashing to address public climate change pressures. Thus, we propose the following hypotheses:

Hypothesis 5a. Uncertainty avoidance negatively moderates the relationship between MNCs' environmental scores and absolute carbon emissions.

Hypothesis 5b. Uncertainty avoidance negatively moderates the relationship between MNCs' environmental scores and carbon intensity.

2.2.5 | Long-term orientation

Companies' long-term strategies may be enhanced in societies with a long-term orientation. Hofstede et al. (2010) stated that long-term orientation measures the extent to which individuals and social norms are exhibited in future values. Compared to short-term orientation societies, individuals' behavioral principles in the long-term orientation context may focus more on long-term goal achievement (Hofstede et al., 2010), requiring them to enhance their abilities to adapt to changes and persist for success (Geletkanycz, 1997). This can lead managers in such societies to be more concerned with stakeholder demands (Cheng et al., 2014) and make adjustments or changes to ensure business success (Geletkanycz, 1997). Therefore, companies tend to adopt social-awareness activities and establish long-term strategies (Miska et al., 2018).

Considering the aforementioned points, corporate ethical decision-making regarding climate change may be strengthened in societies with a long-term orientation. In such societies, managers are normally more concerned about corporate reputation and relationships with outsiders (Hofstede et al., 2010) to maintain long-term planning (Luo & Tang, 2022), which may enhance companies' awareness of stakeholder pressure for business success. Studies have demonstrated that companies from long-term-oriented societies are more willing to share corporate carbon information with stakeholders (Luo & Tang, 2016), promote green products and clean energy projects (Luo & Tang, 2022), and invest in environmental innovations (Ullah et al., 2022), thus promoting ethical corporate decision-making in climate change. Existing evidence indirectly supports these arguments: Ullah et al. (2022) reported that long-term orientation positively affects corporate environmental innovation, and Luo and Tang (2022) indicated a positive relationship between long-term orientation and corporate carbon performance. Other studies also found that long-term orientation can positively impact environmental investment (Ullah et al., 2022) and CSR-related practices (Gallen & Peraita, 2018; García-Sánchez et al., 2016).

Therefore, we follow the arguments of Geletkanycz (1997), Hofstede et al. (2010), and Miska et al. (2018) and assume that managers' social awareness is strengthened in societies with a long-term orientation. This may enhance managers' concerns regarding the ethical harm of business scandal risk, leading to a more serious treatment of stakeholders' climate change pressure. Consequently, the possibility of managers' participation in carbon reduction may increase, and the adoption of a greenwashing strategy may be reduced. Thus, we propose the following hypotheses:

Hypothesis 6a. Long-term orientation negatively moderates the relationship between MNCs' environmental scores and absolute carbon emissions.

Hypothesis 6b. Long-term orientation negatively moderates the relationship between MNCs' environmental scores and carbon intensity.

2.2.6 | Indulgence

Companies' concerns about long-term business success are likely to increase in low-indulgence societies. Hofstede et al. (2010) posit that individuals' natural desires and motivations can be enhanced by increasing indulgence, whereas social norms in self-control may be reduced. Thus, individuals from high-indulgent societies tend to enjoy their lives with the motivation of "having fun" (Pinkse & Kolk, 2012) and are less tolerant of being regulated (Disli et al., 2016). Owing to these influences, companies' attention to economic and public concerns differs. Specifically, companies from low-indulgence societies tend to have more incentives to participate in environmental sustainability-related activities (Gallego-Álvarez & Ortas, 2017) and focus more on public climate change concerns than those with high indulgence (Ullah et al., 2022). Consequently, companies may be more concerned with long-term business success.

Ethical corporate decision-making in climate change may be strengthened in low-indulgence societies. As mentioned, companies from low-indulgence societies may exhibit a higher possibility of prioritizing long-term business success as the priority goal. Therefore, with increasing public awareness, companies must invest more in environmental innovation (Ullah et al., 2022) and increase their CSR transparency for stakeholders (Gallen & Peraita, 2018) to achieve long-term goals. These factors may enhance ethical decision-making regarding climate change. Previous evidence has also indirectly supported this argument: Disli et al. (2016) posited a positive relationship between indulgence and carbon emissions, Gallego-Álvarez and Ortas (2017) indicated that indulgence negatively affects companies' CSR practices, and Ullah et al. (2022) confirmed that indulgence is negatively related to corporate environmental innovations.

Therefore, we follow Hofstede et al. (2010), Gallego-Álvarez and Ortas (2017), and Ullah et al. (2022) and assume that companies' incentives for environment-related activities and investments are likely to encourage low-indulgence societies. This may cause managers to focus more on actual carbon reduction actions than exaggerated "talk" (e.g., greenwashing strategy). Thus, we propose the following hypotheses:

Hypothesis 7a. Indulgence positively moderates the relationship between MNCs' environmental scores and absolute carbon emissions.

Hypothesis 7b. Indulgence positively moderates the relationship between MNCs' environmental scores and carbon intensity.

3 | METHODOLOGY

3.1 | Sample and data

This study focuses on MNCs listed on the 2021 Fortune Global 500 list (between 2001 and 2020) as a sample source, similar to

existing corporate environmental responsibility and related studies (Hussain et al., 2018). We collected carbon emissions data, environmental scores, firms' financial variables, and corporate governance variables from Refinitiv Eikon Asset4³ (Benlemlih & Yavaş, 2023; Shaukat et al., 2016) and matched them with firms from Fortune Global 500 (Hussain et al., 2018). Six national cultural indicators were obtained from Hofstede et al. (2010).

For the sample selection process, we followed the approaches of Hussain et al. (2018) and Peng and Zhang (2022) to remove observations with missing variables in our regression model and exclude the financial industry (SIC 6000–6999). Our final dataset comprised 1403 observations from 199 firms in 27 countries from 2001 to 2020.

3.2 | Variable development and model specifications

3.2.1 | Variable development

For the dependent variables, we refer to Choi and Luo (2020) and Luo and Tang (2022), in which the measurement of corporate carbon reduction must consider absolute emissions and intensity reduction. Therefore, this study selects MNCs' carbon emissions (CE) and carbon intensities (CI) as dependent variables. Specifically, we borrowed the approach of Choi and Luo (2020) to collect the absolute carbon emissions data (i.e., both Scopes 1 and 2) of sample companies from Asset4 for the measurement of CE, whereas CI is equal to the total carbon emissions over net sales (Luo & Tang, 2022),⁴ which were also collected from Asset4.

MNCs' environmental score (EN) is the independent variable. We followed the approach of Shaukat et al. (2016), which used companies' total environmental scores from Refinitiv Eikon Asset4 as the measurement.

National culture dimensions were the moderator variables. Tan and Wang (2011) argue that MNCs' responses to social responsibility in both parent and host countries are part of their global strategy, which is determined by the headquarters' decision-making (i.e., parent country). Considering the cultural influence on managers' moral reasoning, MNCs' carbon reduction decision-making is likely correlated with the parent country's national cultural characteristics. Therefore, we followed the approach of Gallego-Álvarez and Ortas (2017) and Ullah et al. (2022) to adopt the concept of six national cultural dimensions (power distance [PD], individualism [INDI], masculinity [MAS], uncertainty avoidance [UA], long-term orientation [LTO], and indulgence [INDU]) from Hofstede et al. (2010) and used the same measurement results (i.e., parent country).

³Refinitiv Eikon Asset4 (2018–present) formerly known as Thomson Reuters's Asset4; the main changes in Asset4 are that the indicators of economic dimension have been integrated into the environment, society, and governance in 2018. However, the rating results are reliable with comparability whether before or after 2018. Therefore, the latest study has selected the Refinitiv Eikon Asset4 as a data source (Benlemlih & Yavaş, 2023).

⁴We used carbon intensity to control for the inflation of carbon emissions and ensure that the carbon emission data are comparable among different companies and years.

We controlled for several board- and firm-specific characteristics. Previous studies have argued that companies' environmental practices are determined by board strategy decision-making (Peng & Zhang, 2022) and constrained by available resources (Katmon et al., 2019). Therefore, we considered the existing evidence that companies' CSR-related practices may be affected by the audit committee (AUDIT; Hussain et al., 2018), board independence (INDE; Cui et al., 2020; Kaymak & Bektas, 2017; Peng & Zhang, 2022), board meeting frequency per year (BM; Kaymak & Bektas, 2017; Peng & Zhang, 2022), board size (BOARD; Peng & Zhang, 2022), CEO duality (DUALITY; Kaymak & Bektas, 2017; Peng & Zhang, 2022), firm size (TA; Luo, 2019), leverage (LEV; Luo, 2019), market value (MV; Kurt & Peng, 2021), return on equity (ROE; Zhang et al., 2021), and tangibility (TAN; Cheung & Pok, 2019) to adopt the same measurement and select these board- (AUDIT, INDE, BM, BOARD, and DUALITY) and firm-specific (TA, LEV, MV, ROE, and TAN) characteristics as control variables for this study (see Table 1 for descriptions of the variables).

3.2.2 | Model specifications

To explore the relationship between environmental scores and carbon emissions (Hypothesis 1a) and between environmental scores and carbon intensity (Hypothesis 1b), we used the following equations:

$$CE_{t+1} = \beta_0 + \beta_1 EN_t + \text{Control Variables} + \text{Year FE} + \text{Country FE} + \varepsilon, \quad (1)$$

$$CI_{t+1} = \beta_0 + \beta_1 EN_t + \text{Control Variables} + \text{Year FE} + \text{Country FE} + \varepsilon, \quad (2)$$

where *CE* refers to the natural logarithm of the CO₂ Scope 1 and Scope 2 emissions in metric tons, *CI* refers to the natural logarithm of the CO₂ Scope 1 and Scope 2 emissions in metric tons over net sales, *EN* refers to the natural logarithm of the environmental scores, and *Control Variables* refer to the board (AUDIT, INDE, BM, BOARD, and DUALITY) and firm-specific (TA, LEV, MV, ROE, and TAN) control variables. ε refers to error terms. *CE* and *CI* are in period $t+1$ period, whereas *EN* is in period t because the impact of the environmental

TABLE 1 Variable definitions.

Variable	Definition
Dependent variable	
CE	Natural logarithm of yearly CO ₂ Scope 1 and Scope 2 emissions in tons
CI	Natural logarithm of yearly CO ₂ Scope 1 and Scope 2 emissions in tons over (net) sales
Independent variable	
EN	Natural logarithm of environmental performance scores
Moderating variable	
Culture	Natural logarithm of one of the following national culture indicators
PD	Power distance: Measurement results from Hofstede et al. (2010)
INDI	Individualism: Measurement results from Hofstede et al. (2010)
MAS	Masculinity: Measurement results from Hofstede et al. (2010)
UA	Uncertainty avoidance: Measurement results from Hofstede et al. (2010)
LTO	Long-term orientation: Measurement results from Hofstede et al. (2010)
INDU	Indulgence: Measurement results from Hofstede et al. (2010)
PostParis	Equal to 1 if the observation is from post-Paris Agreement period and 0 otherwise
Control variables	
Firm-specific	
TA	Natural logarithm of the total assets
TAN	Net plant, property, and equipment (PPE) over the total assets
MV	Natural logarithm of the total market value of the company
ROE	Return on total shareholders' equities
LEV	Total debt over total assets
Board-specific	
BOARD	Total number of board members
INDE	Percentage of the independent board members over total board members
BM	Total number of board meetings per year
DUALITY	Equal to 1 if the CEO and chairman are the same people and 0 otherwise
AUDIT	Equal to 1 if the company has an audit board committee

score on carbon emissions will likely take some time to take effect. All non-ratio variables are in US dollars (millions), and the year (*Year FE*) and country fixed effects (*Country FE*) are included. All continuous variables are winsorized at the 1% and 99% levels to mitigate the potential influence of outliers.

To examine the moderating effect of national culture (Hypotheses 2a–7b), we employed the following equations:

$$CE_{t+1} = \beta_0 + \beta_1 EN_t + \beta_2 Culture + \beta_3 EN_t \times Culture + Control Variables + Year FE + Country FE + \varepsilon, \quad (3)$$

$$CI_{t+1} = \beta_0 + \beta_1 EN_t + \beta_2 Culture + \beta_3 EN_t \times Culture + Control Variables + Year FE + Country FE + \varepsilon, \quad (4)$$

where *Culture* refers to one of the following national cultural indicators (log form): *PD*, *INDI*, *MAS*, *UA*, *LTO*, or *INDU*.

TABLE 2 Carbon emissions and environmental scores by country.

Country	N	Absolute carbon emission (kmt)	Environmental scores (0–100)
Australia	24	23,508.00	76.59
Austria	11	12,201.27	66.94
Belgium	6	5570.00	56.75
Brazil	1	14,000.00	79.04
Canada	37	8405.28	67.52
China	54	13,854.62	54.15
Denmark	3	37,186.33	74.20
Finland	16	345.58	78.99
France	80	11,455.23	78.40
Germany	98	10,291.43	77.54
India	16	3933.43	76.43
Ireland	44	5319.30	75.52
Italy	4	87,533.67	88.70
Japan	170	6036.39	75.04
Luxembourg	10	139,000.00	84.88
Malaysia	5	6400.00	35.63
Netherlands	48	22,200.66	67.74
Norway	7	14,909.29	78.82
Singapore	4	4240.95	48.28
South Korea	22	23,993.79	77.50
Spain	35	16,977.11	84.08
Sweden	5	392.00	86.38
Switzerland	54	5734.03	80.72
Taiwan	22	363.30	50.76
Thailand	6	53,918.90	93.14
The United Kingdom	123	15,226.60	72.75
The United States	498	10,694.07	70.89
Total	1403		

4 | RESULTS

4.1 | Sample distribution, descriptive statistics, and correlation analysis

Table 2 reports the average carbon emissions and environmental scores as well as sample distribution by country. For the sample distribution, approximately 35.5% (498 of 1403), 12.1% (170 of 1403), and 8.8% (123 of 1403) of the observations are from the United States, Japan, and the United Kingdom, respectively. Conversely, Belgium, Brazil, Denmark, Italy, Malaysia, Norway, Singapore, Sweden, and Thailand have less than 10 observations. For the carbon emissions and environmental performance scores, Luxembourg has the highest average carbon emissions with 139,000.00 thousand metric tons (kmt), followed by Italy (87,533.67 kmt) and Thailand (53,918.90 kmt), while the average carbon emissions in Finland (345.58 kmt), Sweden (392.00 kmt), and Taiwan (363.30 kmt) are lower than 400.00 kmt. In terms of environmental scores, Italy (88.70), Sweden (86.38), and Thailand (93.14) have extremely high average *EN* (over 85 of 100), while Malaysia (35.63) and Singapore (48.28) have scores less than 50.



TABLE 3 Variable statistics.

Variable	Mean	SD	Min	P25	P50	P75	Max
CE	14.81	1.83	10.57	13.62	14.70	15.91	18.83
CI	4.21	2.16	-0.17	2.93	3.77	5.40	13.01
EN	4.25	0.29	2.88	4.13	4.34	4.45	4.57
PD	3.76	0.27	3.26	3.56	3.69	3.99	4.38
INDI	4.20	0.44	2.83	4.09	4.38	4.51	4.51
MAS	4.07	0.38	2.64	4.13	4.13	4.19	4.55
UA	3.98	0.33	3.40	3.83	3.83	4.23	4.52
LTO	3.82	0.52	3.04	3.26	3.93	4.42	4.61
INDU	4.01	0.29	3.18	3.74	4.22	4.22	4.26
TA	11.88	2.04	8.70	10.46	11.34	12.61	19.00
TAN	0.24	0.24	0.00	0.02	0.16	0.40	0.89
MV	10.49	1.28	5.26	9.81	10.56	11.38	12.69
ROE	0.18	0.18	-0.30	0.08	0.15	0.23	0.90
LEV	0.25	0.14	0.00	0.16	0.23	0.34	0.65
BOARD	12.88	3.41	6.00	11.00	12.00	15.00	24.00
INDE	65.95	26.14	0.00	50.00	75.00	86.67	100.00
BM	9.09	3.69	4.00	6.00	8.00	11.00	25.00

Tables 3 and 4 present the sample variable statistics and Pearson correlations. The first column shows the mnemonics of the variables, followed by means, standard deviations, and correlation coefficients. The correlation between the variables was positive or negative, and two extremely high coefficients were observed. These include *CE* and *CI* (.85) and *INDI* and *INDU* (.81), which are similar to previous studies (Eisend et al., 2016; Peng & Zhang, 2022). To avoid multicollinearity, we adopt the same approach as Eisend et al. (2016) and Peng and Zhang (2022) to include only one dependent variable (i.e., *CE* or *CI*) and one national culture dimension variable (i.e., *PD*, *INDI*, *MAS*, *UA*, *LTO*, or *INDU*) in one regression model. Moreover, the variance inflation factors (VIFs) of all model variables were calculated for an additional multicollinearity check. The results demonstrate that the VIFs are between 1.09 and 2.38, and the mean lies between 1.35 and 1.51, suggesting that our models are free of multicollinearity issues (see Table 5 for the VIF results).

4.2 | Multivariate regression results

4.2.1 | Results of MNCs' greenwashing

Table 6 reports the results from Equations (1) and (2). Column (1) presents the relation between environmental scores and carbon emissions (Hypothesis 1a), Column (3) presents the relation between environmental scores and carbon intensity (Hypothesis 1b), and Columns (2) and (4) present a longer lag time period (i.e., 2 years) for carbon emissions and the intensity, respectively. The results also reveal that the adjusted R^2 is over .4 in all columns, indicating that our models effectively explain the variance in carbon emissions/intensity.

As Table 6 shows, *EN* is positively related to *CE* at the 1% significance level in Column (1), indicating that higher environmental scores result in higher carbon emissions, consistent with the arguments of Ramus and Montiel (2005), Berrone et al. (2017), Siano et al. (2017), and Torelli et al. (2020) and similar to the findings of De Vries et al. (2015) and Seele and Gatti (2017). Our results indicate that the environmental score is a strategic tool for misleading stakeholders. Columns (2) and (4) show similar results, indicating that they are robust. These findings are consistent with our expectation that MNCs' greenwashing for carbon reduction exists among firms, thus supporting Hypotheses 1a and 1b.

4.2.2 | Additional tests for greenwashing

To provide a deeper analysis of MNCs' greenwashing, we conduct several additional tests (i.e., countries with mandatory disclosure policy vs. voluntary disclosure, carbon-intensive vs. non-carbon-intensive industry, and firm cluster) to mitigate the potential bias of our research models, and the results are reported in Table 7. In Panel A of Table 7, we test whether our results differ for countries with mandatory or voluntary disclosures. Columns (1) and (2) show the results for observations in countries with mandatory disclosures during the mandatory reporting period, whereas Columns (3) and (4) show the results for observations in countries without mandatory disclosures. Columns (5) and (6) employ a difference-in-differences (DID) method to test the difference between mandatory and voluntary disclosure. Our results show that *EN* is positive and significant across all specifications, suggesting that higher environmental scores result in higher carbon emissions and intensities in both mandatory and voluntary countries. Additionally, the interaction term $Post_Man \times EN$ is

TABLE 4 Pearson's correlation.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
(1) CE	1																
(2) CI	.85***	1															
(3) EN	.21***	.11***	1														
(4) PD	-.06**	-.03	-.11***	1													
(5) INDI	.10***	.05**	.13***	-.64***	1												
(6) MAS	-.01	.02	.03	.02	-.05*	1											
(7) UA	.00	-.05*	.16***	.35***	-.27***	.09***	1										
(8) LTO	-.09***	-.06**	.01	.38***	-.66***	.04*	.53***	1									
(9) INDU	.07**	.04	.03	-.70***	.81***	-.15***	-.37***	-.72***	1								
(10) TA	.03	-.03	.13***	.47***	-.56***	.24***	.49***	.49***	-.54***	1							
(11) TAN	.46***	.29***	.13***	-.31***	.38***	-.15***	-.18***	-.23***	.33***	-.50***	1						
(12) MV	-.04*	-.38***	.20***	-.17***	.28***	.01	-.09***	-.28***	.23***	-.07***	.15***	1					
(13) ROE	-.14***	-.11***	-.02	-.18***	.21***	.01	-.27***	-.25***	.22***	-.26***	.07**	.21***	1				
(14) LEV	.18***	.12***	.10***	.02	.04	.01	.02	.02	-.01	.06**	.13***	.00	.06**	1			
(15) BOARD	.11***	.05*	.19***	.13***	-.05*	.20***	.33***	.31***	-.32***	.20***	-.01	.01	-.13***	.21***	1		
(16) INDE	.11***	.05*	.07**	-.52***	.56***	-.22***	-.47***	-.67***	.64***	-.52***	.30***	.27***	.21***	-.04	-.34***	1	
(17) BM	-.02	.01	.09***	.16***	-.08**	.10***	.31***	.13***	-.08***	.37***	-.26***	-.13***	-.11***	.02	.02	-.22***	1

Note: The first and second variables are MNCs' absolute carbon emissions (CE) and the intensity (CI), which are the dependent variables. Dependent variables are followed by the independent variables (MNCs' environmental scores [EN]). The independent variable is followed by the moderator variables (national culture concept of six dimensions), which were introduced by Hofstede et al. (2010). National culture dimension variables include power distance (PD), individualism (INDI), masculinity (MAS), uncertainty avoidance (UA), long-term orientation (LTO), and indulgence (INDU). Moderator variables are followed by the control variables, which include firm size (TA), tangibility (TAN), market value (MV), return on equity (ROE), leverage (LEV), board size (BOARD), board independence (INDE), and board meeting frequency per year (BM).

*Statistical significance at 10%.

**Statistical significance at 5%.

***Statistical significance at 1%.

TABLE 5 VIF across all models.

Variable	VIF	VIF	VIF	VIF	VIF	VIF	VIF
EN	1.18	1.20	1.19	1.18	1.19	1.18	1.18
PD		1.53					
INDI			1.98				
MAS				1.11			
UA					1.62		
LTO						2.06	
INDU							2.05
TA	2.02	2.14	2.38	2.05	2.14	2.13	2.22
TAN	1.50	1.51	1.51	1.50	1.51	1.51	1.51
MV	1.20	1.20	1.24	1.20	1.20	1.24	1.21
ROE	1.15	1.15	1.15	1.16	1.17	1.16	1.15
LEV	1.09	1.09	1.10	1.10	1.10	1.10	1.10
BOARD	1.26	1.27	1.30	1.29	1.31	1.28	1.29
INDE	1.67	1.86	1.91	1.68	1.77	2.20	2.03
BM	1.19	1.20	1.26	1.19	1.23	1.21	1.24
Mean VIF	1.36	1.41	1.50	1.35	1.42	1.51	1.50

insignificant in Columns (5) and (6). These results indicate that MNCs' greenwashing in carbon reduction exists in countries with both mandatory and voluntary disclosure.

Panel B of Table 7 reports the results for carbon- and non-carbon-intensive industries, similar to the approach adopted by Choi et al. (2021).⁵ The results show that *EN* is positive and significant in all columns, suggesting that greenwashing exists in both carbon- and non-carbon-intensive industries. To avoid firm-specific problems, we include firm-clustered standard errors in Panel C of Table 7, and the results are stable. Furthermore, we borrowed the approach of Luo and Wu (2019) to mitigate endogeneity using two-stage least squares (2SLS) regressions. Specifically, we include foreign sales as the instrumental variable because this variable is correlated with *EN* but not with our dependent variables. In the first stage, the coefficient of *ForeignSales* is positive and significant, indicating that a higher percentage of foreign sales are expected to be associated with a higher environmental score, which is reasonable because Park (2018) argues that foreign sales are the key element in measuring companies' internationalization. With increased internationalization, the number of host countries involved in companies' operations will also increase. Therefore, companies face more complex stakeholder environmental demands from their host countries. Consequently, companies must strengthen their environmental practices to meet their legitimacy needs, which increases their environmental scores. In the second stage, the conclusions remain unchanged.

⁵We classified agriculture, mining, construction, and transportation as carbon-intensive industries, which is similar to Choi et al. (2021).

4.2.3 | Moderating effects of national culture

Table 8 reports the moderating effect of six national culture dimensions (Hypotheses 2a–7b), respectively. For the moderating effect of national culture, the result of the interaction term $EN \times INDI$ is negative, with a 10% significance level in Column (2), suggesting that the moderating effect of individualism is negative; this supports the arguments and findings of Scholtens and Dam (2007) and Disli et al. (2016). Our results reveal that managers may demonstrate higher ethical standards in high-individualism societies and participate in actual actions in carbon reduction. Considering that the above results are consistent with our expectations, Hypothesis 3a is supported.

The interaction term $EN \times MAS$ is negatively related to *CE* (Column [3]) and *CI* (Column [9]) at the 5% and 10% significance levels, respectively, indicating that the moderating effect of masculinity is negative, which is consistent with the arguments and evidence of Okpara (2014), Torelli et al. (2020), and Ullah et al. (2022). Our results reveal that managers from high-masculinity societies are unlikely to consider unethical decision-making (e.g., greenwashing), which is consistent with our expectations; thus, Hypotheses 4a and 4b are supported.

Moreover, the interaction term $EN \times UA$ is negatively related to *CE* at the 1% significance level in Columns (4) and (10), suggesting that the moderating effect of uncertainty avoidance is negative. Considering previous arguments (Griffin et al., 2017; Li & Harrison, 2008) and evidence (Luo & Tang, 2016; Okpara, 2014), the results indicate that managers from high-uncertainty avoidance societies tend to calculate the risk more cautiously, which may enhance their awareness of business scandal (e.g., greenwashing) risk and encourage them “act” more and “talk” less in carbon reduction. This finding is consistent with our expectations; thus, Hypotheses 5a and 5b were supported.

The results for the interaction terms $EN \times PD$ (Columns [1] and [7]), $EN \times LTO$ (Columns [5] and [11]), and $EN \times INDU$ (Columns [6] and [12]) are insignificant, indicating no moderating effect for power distance, long-term orientation, or indulgence. This is inconsistent with previous arguments (Hofstede et al., 2010; Miska et al., 2018; Ullah et al., 2022), evidence (Disli et al., 2016; Luo & Tang, 2022; Ullah et al., 2022), and our expectations. Thus, Hypotheses 2a, 2b, 6a, 6b, 7a, and 7b are not supported. Meanwhile, the results of $EN \times INDI$ (Column [8]) are also insignificant, indicating that individualism does not have a moderating effect between MNCs' environmental scores and carbon intensity. These results do not confirm previous findings (Disli et al., 2016; Scholtens & Dam, 2007) and our expectations; thus, Hypothesis 3b is unsupported.

4.2.4 | Further analyses: Moderating effects of the Paris Agreement

As one of the most important institutional factors in climate change, the Paris Agreement may affect MNCs' strategic greenwashing behavior. He et al. (2021) stated that the Paris Agreement delivered a clear message to governments and the private sector to decarbonize and

TABLE 6 OLS regression results of MNCs' greenwashing.

Variable	CE_{t+1} (1)	CE_{t+2} (2)	CI_{t+1} (3)	CI_{t+2} (4)
EN_t	0.775*** (5.13)	0.678*** (4.09)	1.304*** (7.09)	1.168*** (5.81)
TA_t	0.418*** (9.65)	0.458*** (9.55)	0.397*** (7.53)	0.468*** (8.03)
TAN_t	4.763*** (21.51)	4.469*** (18.73)	3.354*** (12.43)	3.065*** (10.57)
MV_t	-0.353*** (-10.25)	-0.370*** (-9.93)	-0.995*** (-23.68)	-1.005*** (-22.19)
ROE_t	-0.786*** (-3.41)	-0.650*** (-2.69)	-0.222 (-0.79)	-0.089 (-0.30)
LEV_t	0.923*** (3.21)	0.602* (1.91)	0.609* (1.74)	0.300 (0.78)
$BOARD_t$	0.027* (1.70)	0.026 (1.47)	0.010 (0.53)	-0.003 (-0.12)
$INDE_t$	0.009*** (3.48)	0.009*** (3.18)	0.008*** (2.72)	0.009*** (2.82)
BM_t	-0.004 (-0.34)	-0.008 (-0.57)	-0.012 (-0.77)	-0.020 (-1.18)
$DUALITY_t$	-0.298*** (-3.22)	-0.353*** (-3.51)	-0.409*** (-3.63)	-0.436*** (-3.57)
$AUDIT_t$	0.252 (1.28)	0.252 (1.17)	-0.087 (-0.37)	-0.188 (-0.72)
Constant	9.600*** (9.87)	9.629*** (9.10)	5.333*** (4.50)	5.242*** (4.08)
Year FE	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
N	1403	1170	1403	1170
Adjusted R^2	.446	.440	.426	.431

*Statistical significance at 10%.

**Statistical significance at 5%.

***Statistical significance at 1%.



TABLE 7 Additional analysis.

Panel A: Mandatory disclosure versus voluntary disclosure						
	Mandatory only		Voluntary only		DID for the full sample	
Variable	CE_{t+1} (1)	CI_{t+1} (2)	CE_{t+1} (3)	CI_{t+1} (4)	CE_{t+1} (5)	CI_{t+1} (6)
EN_t	1.532*** (3.73)	1.752*** (3.48)	0.510*** (3.46)	1.090*** (6.07)	0.781*** (5.17)	1.312*** (7.12)
$Post_Man \times EN_t$					−0.048 (−1.17)	−0.060 (−1.19)
All control variables are included						
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
N	289	289	1114	1114	1403	1403
Adjusted R ²	.534	.452	.509	.516	.446	.426
Panel B: Carbon-intensive versus non-carbon-intensive industry						
	Carbon-intensive		Non-carbon-intensive			
Variable	CE_{t+1} (1)	CI_{t+1} (2)	CE_{t+1} (4)	CI_{t+1} (5)		
EN_t	1.197** (2.53)	1.348** (2.38)	0.651*** (4.35)	1.245*** (6.72)		
All control variables are included						
Year FE	Yes	Yes	Yes	Yes		
Country FE	Yes	Yes	Yes	Yes		
N	323	323	1080	1080		
Adjusted R ²	.372	.570	.539	.454		
Panel C: Firm cluster included						
Variable			CE_{t+1} (1)	CI_{t+1} (2)		
EN_t			0.775** (2.56)	1.304*** (3.44)		
All control variables are included						
Year FE			Yes	Yes		
Country FE			Yes	Yes		
N			1403	1403		
Adjusted R ²			.446	.426		
Panel D: Two-stage least squares (2SLS)						
	First stage		Second stage ($EN_t = ForeignSales_t$)			
Variable	EN_t (1)		CE_{t+1} (2)	CI_{t+1} (3)		
$ForeignSales_t$	0.170*** (6.80)					
EN_t			7.363*** (5.83)	11.699*** (6.43)		
All control variables are included						
Year FE	Yes		Yes	Yes		

TABLE 7 (Continued)

Panel D: Two-stage least squares (2SLS)			
Variable	First stage EN_t (1)	Second stage ($EN_t = ForeignSales_t$)	
		CE_{t+1} (2)	CI_{t+1} (3)
Country FE	Yes	Yes	Yes
N	1403	1403	1403
Adjusted R^2	.306	.231	.215

*Statistical significance at 10%.

**Statistical significance at 5%.

***Statistical significance at 1%.

consequently strike a balance between economic development and the climate change crisis. This has pushed policymakers to adjust or introduce climate change-related regulations and increased the institutional pressures undertaken by MNCs, which may prevent their strategic greenwashing behavior. However, few studies have investigated the moderating effects of the Paris Agreement in different national cultural contexts. Therefore, based on our empirical regression models, we introduce the following DID regression equation as follows:

$$CE_{t+1} = \beta_0 + \beta_1 EN_t + \beta_2 Culture + \beta_3 PostParis + \beta_4 EN_t \times Culture \times PostParis + Control Variables + Year FE + Country FE + \varepsilon, \quad (5)$$

$$CI_{t+1} = \beta_0 + \beta_1 EN_t + \beta_2 Culture + \beta_3 PostParis + \beta_4 EN_t \times Culture \times PostParis + Control Variables + Year FE + Country FE + \varepsilon, \quad (6)$$

where *PostParis* is equal to 1 if the observation is from post-Paris Agreement period and 0 otherwise.

Table 9 reports the results of the moderating effect of the Paris Agreement. Overall, the coefficients on the *PostParis* (Columns [1] and [8]) are generally insignificant, indicating that the influence of the Paris Agreement on MNCs' carbon reduction is limited. In terms of the interaction terms, the results of $EN \times MAS \times PostParis$ (Columns [4] and [11]), $EN \times UA \times PostParis$ (Columns [5] and [12]), $EN \times LTO \times PostParis$ (Columns [6] and [13]), $EN \times INDI \times PostParis$ (Column [10]), and $EN \times INDU \times PostParis$ (Column [14]) are insignificant. These indicate that for companies from societies with three national culture characteristics (masculinity, uncertainty avoidance, and long-term orientation), the relationship between their environmental scores and carbon emissions is the same before and after the Paris Agreement, and the relationship between their environmental scores and carbon intensity is the same before and after the Paris Agreement for the companies from societies with national culture characteristics of individualism, masculinity, uncertainty avoidance, long-term orientation, and indulgence.

Moreover, the results of $EN \times INDI \times PostParis$ (Column [3]) and $EN \times INDU \times PostParis$ (Column [7]) are negative at the 10% and 5% significance levels, respectively, indicating that the Paris Agreement has effectively prevented companies' strategic greenwashing behavior in societies characterized by individualism or indulgence. Meanwhile,

the interaction term $EN \times PD \times PostParis$ is positively correlated with *CE* (Column [2]) and *CI* (Column [9]) at the 10% significance level. This remarkable result demonstrates that after the Paris Agreement, managers' strategic greenwashing behavior has worsened in societies characterized by high-power distance, which policymakers should be aware of.

4.3 | Robustness tests

Three tests were conducted to evaluate the empirical results. First, previous studies have suggested that ROE and return on assets (ROA) are acceptable financial performance measurements (Barauskaite & Streimikiene, 2021; Barnett & Salomon, 2012). Therefore, we empirically tested the regression models with ROA substituted by ROE for Robustness Test 1 (see Panel A of Table 10 for the results). Thereafter, we drop countries with fewer than 10 observations for Robustness Test 2, similar to Christensen et al. (2013) (see Panel B of Table 10 for results). Third, we use alternative carbon intensity (i.e., total emissions over the cost of goods sold [COGS]) as a dependent variable (Downar et al., 2021) (see Panel C of Table 10 for results).

The results of the robustness tests are similar to our empirical results, meaning that Hypotheses 1a, 1b, 3a, 4a, 4b, 5a, and 5b pass the robustness tests.

5 | DISCUSSION

Over the past few decades, increasing public climate change concerns have generated significant pressure on MNCs for carbon reduction, which requires companies to participate in related actions to meet their legitimate needs. Meanwhile, academic attention to companies' strategic greenwashing behavior is also emerging, particularly regarding the issue of misleading stakeholders. Under these circumstances, managers adopting environmental scores to mislead stakeholders will not only generate deceptive manipulation, thus slowing down the global progress of climate change pressure relief, but will also become an academic concern worthy of in-depth investigation. However,



TABLE 8 OLS regression results of national cultural moderating effects.

Variable	CE _{t+1}				CI _{t+1}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
EN _t	0.782 (0.60)	2.223*** (2.71)	2.299*** (3.31)	4.373*** (3.38)	0.288 (0.44)	2.383*** (2.14)	2.778* (1.75)	2.035** (2.04)	2.879*** (3.40)	6.591*** (4.19)	1.760*** (2.20)	1.542 (1.13)
PD	-14.392** (-2.37)						-16.685** (-2.26)					
EN _t × PD	-0.002 (-0.01)						-0.386 (-0.93)					
INDI		-65.068** (-2.40)						-84.805** (-2.56)				
EN _t × INDI		-0.334* (-1.80)						-0.169 (-0.74)				
MAS			-45.677** (-2.48)						-58.499*** (-2.60)			
EN _t × MAS			-0.366** (-2.25)						-0.378* (-1.90)			
UA				10.995*** (3.43)						14.814*** (3.79)		
EN _t × UA				-0.899*** (-2.80)						-1.321*** (-3.38)		
LTO					-4.036** (-2.53)						-3.893** (-2.00)	
EN _t × LTO					0.125 (0.76)						-0.117 (-0.58)	
INDU						18.735*** (2.66)						22.175*** (2.58)
EN _t × INDU						-0.397 (-1.46)						-0.059 (-0.18)

TABLE 8 (Continued)

Variable	CE_{t+1}					CI_{t+1}						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
All control variables are included												
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1403	1403	1403	1403	1403	1403	1403	1403	1403	1403	1403	1403
Adjusted R ²	.447	.448	.449	.446	.447	.447	.426	.426	.427	.430	.426	.426

*Statistical significance at 10%.

**Statistical significance at 5%.

***Statistical significance at 1%.

research on this topic is lacking. Therefore, this study focused on MNCs featured in the Fortune Global 500 lists between 2001 and 2020 to investigate the relationship between environmental scores, carbon emissions/intensity, and the moderating effect of national culture.

5.1 | Findings

Following the investigation, several valuable findings were obtained. First, the environmental score is positively related to MNCs' carbon emissions (Hypothesis 1a) and intensity (Hypothesis 1b), indicating that companies are successful in strengthening their environmental statements, while failing to reduce carbon emissions and intensity. This is a typically “talk” more and “act” less behavior of greenwashing, as suggested by Siano et al. (2017). More specifically, a high environmental score is formed into an impression management tool for misleading stakeholder perceptions, which allows managers to strategically adopt these tools as greenwashing to windowdress their carbon reduction. The reason for this phenomenon is the lack of mandatory third-party auditing requirements for voluntary corporate disclosure (Ramus & Montiel, 2005) and no financial penalties for companies unable to fulfill their environmental commitment (Berrone et al., 2017), which creates opportunities for managers to strategically exaggerate the environmental statement for legitimacy strategic needs (Lyon & Montgomery, 2015). In particular, with the increasing global climate change concerns, the above regulation may become the “trigger” for MNC managers to strategically mislead stakeholders by enhancing environmental statements to cover their failure at carbon reduction. Meanwhile, our additional test results show that, regardless of whether the companies are from carbon-intensive or non-intensive industries, or whether these are countries with mandatory or voluntary disclosure regulations, greenwashing in carbon reduction remains the same. These findings support our arguments, and MNCs' greenwashing in climate change (i.e., window-dressing carbon emissions and intensity) was confirmed.

Second, individualism negatively moderates the relationship between the environmental score and carbon emissions (Hypothesis 3a), which indicates that with an increased environmental score, the carbon emissions of MNCs from high-individualism societies are lower than those from low-individualism societies. This is because stakeholder demands are enhanced in highly individualistic societies (Husted, 2005), which increases managers' awareness of ethical policies (Scholtens & Dam, 2007) and environmental accountability (Disli et al., 2016). Thus, managers should focus more on reducing corporate carbon emissions. Moreover, the moderating effects of masculinity (Hypotheses 4a and 4b) are negative. These findings reveal that with an increased environmental score, MNCs from high-masculinity societies exhibit more carbon reduction actions than those from low-masculinity societies. In particular, compared with low-masculinity societies, managers from high-masculinity societies tend to show a stronger sense of success desire (Ullah et al., 2022) and higher moral values (Okpara, 2014), which causes them to be more concerned

TABLE 9 Further analysis (the Paris Agreement).

Variable	CF_{t+1}									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
EN_t	0.775*** (5.13)	0.645*** (3.89)	0.859*** (5.41)	0.767*** (4.82)	0.739*** (4.57)	0.718*** (4.60)	0.907*** (5.51)	1.304*** (7.09)	1.140*** (5.64)	1.352*** (6.98)
$PostParis$	-0.816 (-1.30)	-2.254** (-2.28)	0.266 (0.30)	-0.908 (-1.02)	-1.235 (-1.35)	-1.528* (-1.90)	0.777 (0.77)	-1.112 (-1.46)	-2.928** (-2.43)	-0.501 (-0.46)
PD		-14.809** (-2.53)							-18.979*** (-2.66)	
$EN_t \times PD \times PostParis$		0.092* (1.88)							0.116* (1.94)	
$INDI$			-65.394** (-2.41)							-84.882** (-2.56)
$EN_t \times INDI \times PostParis$			-0.061* (-1.72)							-0.035 (-0.80)
MAS				-45.543** (-2.46)						
$EN_t \times MAS \times PostParis$				0.005 (0.15)						
UA					7.191** (2.47)					
$EN_t \times UA \times PostParis$					0.025 (0.63)					
LTO						-3.536** (-2.51)				
$EN_t \times LTO \times PostParis$						0.045 (1.41)				
$INDU$							16.816** (2.42)			
$EN_t \times INDU \times PostParis$							-0.095** (-2.01)			

TABLE 9 (Continued)

Variable	CE _{t+1}			CI _{t+1}						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
All control variables are included										
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1403	1403	1403	1403	1403	1403	1403	1403	1403	1403
Adjusted R ²	.446	.447	.447	.446	.446	.447	.448	.426	.427	.426

*Statistical significance at 10%.
**Statistical significance at 5%.
***Statistical significance at 1%.

TABLE 9 (Continued)

Variable	CI _{t+1}		(13)		(14)	
	(11)	(12)	(13)	(14)	(15)	(16)
EN _t	1.291*** (6.65)	1.239*** (6.29)	1.239*** (6.51)	1.399*** (6.96)		
PostParis	−1.271 (−1.18)	−1.875* (−1.68)	−1.940** (−1.98)	0.023 (0.02)		
PD						
EN _t × PD × PostParis						
INDI						
EN _t × INDI × PostParis						
MAS	−58.443*** (−2.59)					
EN _t × MAS × PostParis	0.009 (0.21)					
UA		9.236*** (2.60)				
EN _t × UA × PostParis		0.046				

TABLE 9 (Continued)

Variable	CI_{t+1}		
	(11)	(12)	(13)
		(0.94)	(14)
LTO			
			−4.524*** (−2.64)
$EN_t \times LTO \times PostParis$			0.052 (1.35)
INDU			21.726** (2.56)
$EN_t \times INDU \times PostParis$			−0.067 (−1.17)
All control variables are included			
Year FE	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
N	1403	1403	1403
Adjusted R^2	.426	.426	.426

*Statistical significance at 10%.

**Statistical significance at 5%.

***Statistical significance at 1%.

TABLE 10 Robustness tests results of national cultural moderating effects.

Panel A: Alternative control variable (using ROA instead of ROE)										
Variable	CE_{t+1}				CI_{t+1}					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
EN_t	0.729*** (4.79)	0.536 (0.41)	2.271*** (2.76)	2.405*** (3.46)	3.985*** (3.07)	0.133 (0.20)	2.459** (2.20)	1.145*** (6.31)	2.185 (1.40)	2.095** (2.14)
PD		−12.529** (−2.06)							−12.442* (−1.72)	
$EN_t \times PD$		0.051 (0.15)							−0.273 (−0.67)	
INDI			−55.110** (−2.02)							−62.323* (−1.92)
$EN_t \times INDI$			−0.356* (−1.91)							−0.219 (−0.99)
MAS				−38.983** (−2.11)						
$EN_t \times MAS$				−0.403** (−2.47)						
UA					9.593*** (2.97)					
$EN_t \times UA$					−0.813** (−2.52)					
LTO						−3.656** (−2.28)				
$EN_t \times LTO$						0.153 (0.93)				
INDU							16.344** (2.31)			
$EN_t \times INDU$							−0.427 (−1.56)			
ROA _t	0.020** (2.29)	0.020** (2.29)	0.020** (2.34)	0.021** (2.40)	0.019** (2.17)	0.020** (2.33)	0.020** (2.33)	0.543*** (10.17)	0.542*** (10.15)	0.542*** (10.16)



TABLE 10 (Continued)

Panel A: Alternative control variable (using ROA instead of ROE)									
Variable	CE_{t+1}				CI_{t+1}				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(10)
All control variables are included									
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1402	1402	1402	1402	1402	1402	1402	1402	1402
Adjusted R^2	.444	.443	.445	.446	.446	.444	.444	.450	.450

TABLE 10 (Continued)

Panel A: Alternative control variable (using ROA instead of ROE)				
Variable	CI_{t+1}			
	(11)	(12)	(13)	(14)
EN_t	3.011*** (3.63)	5.828*** (3.77)	1.315* (1.67)	1.676 (1.26)
PD				
$EN_t \times PD$				
INDI				
$EN_t \times INDI$				
MAS	-43.390** (-1.97)			
$EN_t \times MAS$	-0.449** (-2.31)			
UA		11.810*** (3.08)		
$EN_t \times UA$		-1.169*** (-3.05)		

TABLE 10 (Continued)

Panel A: Alternative control variable (using ROA instead of ROE)														
Variable	CI _{t+1}													
	(11)	(12)	(13)	(14)										
LTO			−3.087 (−1.62)											
EN _t × LTO			−0.044 (−0.22)											
INDU				16.794** (1.99)										
EN _t × INDU				−0.131 (−0.40)										
ROA _t		0.548*** (10.28)	0.543*** (10.17)	0.541*** (10.13)										
All control variables are included														
Year FE	Yes	Yes	Yes	Yes										
Country FE	Yes	Yes	Yes	Yes										
N	1402	1402	1402	1402										
Adjusted R ²		.451	.453	.449										
Panel B: Drop countries with observations of less than 10														
Variable	CE _{t+1}													
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
EN _t	0.776*** (5.07)	0.752 (0.57)	2.249*** (2.70)	2.311*** (3.28)	4.470*** (3.38)	0.290 (0.44)	2.383** (2.11)	1.307*** (7.01)	2.742* (1.70)	2.067** (2.04)	2.877*** (3.35)	6.730*** (4.19)	1.763** (2.17)	1.543 (1.12)
PD		−14.433** (−2.35)							−16.676** (−2.23)					
EN _t × PD		0.006 (0.02)							−0.376 (−0.89)					
INDI			−65.050** (−2.37)							−84.502** (−2.52)				
EN _t × INDI			−0.340* (−1.80)							−0.175 (−0.76)				
MAS				−45.744** (−2.45)							−58.372** (−2.57)			

(Continues)

TABLE 10 (Continued)

Panel B: Drop countries with observations of less than 10

Variable	CE _{t+1}				CI _{t+1}									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
EN _t × MAS				−0.369** (−2.23)							−0.377* (−1.87)			
UA					11.099*** (3.41)							14.931*** (3.77)		
EN _t × UA					−0.923*** (−2.81)							−1.355*** (−3.40)		
LTO						−4.035** (−2.49)							−3.880** (−1.97)	
EN _t × LTO						0.125 (0.75)							−0.117 (−0.58)	
INDU							18.736*** (2.63)							22.105** (2.54)
EN _t × INDU							−0.397 (−1.44)							−0.058 (−0.17)
All control variables are included														
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1362	1362	1362	1362	1362	1362	1362	1362	1362	1362	1362	1362	1362	1362
Adjusted R ²	.428	.428	.429	.430	.431	.428	.429	.414	.414	.414	.415	.419	.414	.414
Panel C: Alternative dependent variable (using CI _{t+1} [over COGS] instead of CI _{t+1} [over sales])														
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)							
EN _t	1.194*** (6.68)	4.243* (1.81)	0.513 (0.48)	3.452*** (4.06)	4.205*** (2.63)	2.050** (2.53)	−1.345 (−0.85)							
PD		−14.529** (−1.99)												
EN _t × PD		−0.808 (−1.30)												
INDI			−85.539*** (−2.74)											

TABLE 10 (Continued)

Panel C: Alternative dependent variable (using CI_{t+1} [over COGS] instead of CI_{t+1} [over sales])							
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$EN_t \times INDI$			0.155 (0.65)				
MAS				−57.839*** (−2.74)			
$EN_t \times MAS$				−0.544*** (−2.72)			
UA					12.240*** (3.28)		
$EN_t \times UA$					−0.748* (−1.89)		
LTO						−3.348* (−1.79)	
$EN_t \times LTO$						−0.224 (−1.09)	
INDU							19.135** (2.35)
$EN_t \times INDU$							0.620 (1.61)
All control variables are included							
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1339	1339	1339	1339	1339	1339	1339
Adjusted R ²	.449	.449	.448	.451	.450	.449	.449

*Statistical significance at 10%.

**Statistical significance at 5%.

***Statistical significance at 1%.

about stakeholders' perceptions of climate change for long-term business success achievement and less likely to consider unethical decision-making (i.e., greenwashing) for short-term financial interests. Meanwhile, the moderating effects of uncertainty avoidance (Hypotheses 5a and 5b) are also negative, indicating that compared with the low-uncertainty avoidance societies, MNCs from high-uncertainty avoidance societies are more balanced between "talk" and "act" in carbon reduction. This is because, under the influence of a high-uncertainty avoidance culture, managers' risk calculation is strengthened (Hofstede et al., 2010), which encourages them to follow the rules (Griffin et al., 2017) and prefer a detailed and structured management style (Li & Harrison, 2008) to deal with stakeholders' carbon reduction demands with balanced practices and disclosures. Consequently, the possibility of managers adopting strategic greenwashing for carbon reduction is reduced in societies with high-uncertainty avoidance.

Third, our study reported some unexpected results, including the moderating effects of power distance (Hypotheses 2a and 2b), long-term orientation (Hypotheses 6a and 6b), and indulgence (Hypotheses 7a and 7b), all of which were insignificant. Our hypotheses are based on previous arguments and align with the theory. Unfortunately, they were not supported by our results. We suspect that managers' ethical decision-making may be affected by high or low cultural characteristics differently, especially in societies with power distance, long-term orientation, and indulgence. For example, we assumed that a centralized management style in high-power distance societies (Hofstede et al., 2010) might generate autocratic power authorities (Rowley & Bae, 2002), which would allow them to hold decision-making power and control board nominations (Baldenius et al., 2014). This reduces subordinates' doubts (Weaver, 2001) and encourages top management to adopt exaggerated environmental statements to mislead stakeholders. However, if these powerful authorities are concerned about climate change, they may nominate board candidates with moral values similar to their own. Consequently, companies may focus more on actual action in carbon reduction rather than just "talk." Meanwhile, the moderating effect of individualism between the environmental score and carbon intensity is insignificant (Hypothesis 3b); this means that MNCs' carbon intensity does not differ much between high- and low-individualism societies. Considering this result with our finding of Hypothesis 3a, the influence of individualism on managers' "talk" and "act" in carbon reduction is more complex than our expectation. Previous studies have suggested that the influence of power distance on individuals interacts with individualism (Hofstede et al., 2010; Minkov, 2018), which may explain this phenomenon. Future studies from a leadership perspective may address this issue.

In terms of long-term orientation, companies from such societies exhibit high social awareness (Miska et al., 2018) and ethical harm concern level (Geletkanycz, 1997), which may reduce the possibility of managers adopting greenwashing strategies. However, short-term orientation characteristics emphasize the value of the present for both individuals and organizations (Hofstede et al., 2010). This means that if stakeholders' climate change concerns increase, companies may strengthen their participation in carbon reduction to meet their

present legitimacy needs. Similarly, individuals' natural desires are enhanced in high-indulgence societies (Hofstede et al., 2010). Consequently, MNCs are forced to reduce carbon emissions by increasing stakeholder carbon reduction demands in high-indulgence societies due to global warming issues. Further investigation from a stakeholder perspective helps explain the roles of long-term orientation and indulgence.

Notable findings were obtained from the further analyses. The results show that after the Paris Agreement, there was a positive relationship between MNCs' environmental scores and carbon emissions/intensity in high-power distance societies. This finding is consistent with our expectations regarding the moderating effects of power distance. As mentioned before, the companies' greenwashing may be triggered by the lack of regulations; furthermore, it may even be caused by the institutional pressures generated by the failure of the Paris Agreement to prevent MNCs' misleading behaviors and have only encouraged them to exaggerate the environmental statement for seeking legitimacy. Meanwhile, the Paris Agreement prevents MNCs from adopting environmental scores to cover increasing carbon emissions in societies with individualism or indulgence in cultural characteristics. This finding confirms that individuals' responsibilities (Disli et al., 2016) and companies' environmental accountability (Husted, 2005) are enhanced in societies with high individualism. With the increasing institutional pressure after the Paris Agreement, MNCs may focus more on carbon reduction actions in such societies. Moreover, as in the discussion on the insignificant moderating effect of indulgence, high-indulgence cultural characteristics may increase stakeholders' demand for carbon reduction. Thus, the increasing institutional pressure from the Paris Agreement encourages MNCs to reduce carbon emissions in societies having high-indulgence characteristics.

5.2 | Contributions

Our study discussed how and why MNCs' environmental scores become strategic greenwashing tools to address carbon reduction by considering the influence of national culture from the perspective of institutional theory, which offers the following contributions. First, we explain theoretically the existing greenwashing and greenwashing in covering the failure of carbon emission reduction, and it is the motivation that emphasizes that the driver of MNCs' greenwashing in carbon reduction is relieving public climate change pressures for legitimacy purposes. This differs from previous studies (Clarkson et al., 2008, 2015), in which Clarkson et al. (2008) investigated the consistency between corporate environmental performance and its disclosure, and Clarkson et al. (2015) focused on how institutional factors (i.e., carbon emission trading schemes) lead to the valuation of corporate carbon emissions. However, we focused on a timely and urgent issue of global climate change to analyze the balance between companies' "talk" (environmental scores) and their "actions" (carbon emissions/intensity). These not only introduced a new form of greenwashing in which MNCs have adopted environmental scores to mislead stakeholders in covering their increased carbon emissions and

intensity but also revealed that the reason for this greenwashing is the lack of regulation and the companies' profit desires.

Subsequently, we investigated the influence of national culture, explaining the role of different national cultural dimensions in companies' greenwashing, emphasizing that managers' ethical decision-making in diverse cultural societies varies. These theories explain how managers' ethical decision-making is interrupted by their domestic cultural characteristics, leading to diverse decision-making in the balance between talk and action with reliable empirical evidence. Our study established a link between national culture and greenwashing fields from an institutional theory perspective and extended existing national culture studies (Manta et al., 2022; Vitolla et al., 2019) by considering culture as an institutional factor at the macro-level to both macro considerations (i.e., institutional pressure on MNCs) and micro considerations (i.e., influence on managers' moral reasoning). This extended the application of institutional theory in the interdisciplinary study of greenwashing and management, enriched the field literature, and provided a new research inspiration for future studies.

Finally, further analyses of the Paris Agreement's moderating effects emphasized that the institutional pressure generated by the Paris Agreement did not prevent MNCs from greenwashing in carbon reduction, indicating the powerlessness of the agreement, especially in societies with high-power distance characteristics. The institutional pressure of the agreement worsens managers' "talk" more and leads to less "act." Based on our knowledge, this is the first evidence of an investigation into MNCs' greenwashing in carbon reduction, which not only closes the gap and deepens our understanding of the managers' motivation for such greenwashing behavior and how national culture can influence it but also indicates the directions for academics and precautions to be aware of the MNCs' greenwashing behavior with the consideration of national culture.

5.3 | Implications for policymakers

This study provided suggestions for policymakers. Our results indicate that MNCs adopt environmental scores without actual carbon reduction actions to mislead stakeholders, and the results remain the same after considering whether the industry is carbon-intensive and whether countries have mandatory disclosure regulations. Based on our theoretical discussion and results, we suggest that policymakers realize this new form of greenwashing and establish an independent audit system for nonfinancial information disclosure, especially corporate carbon disclosure. A financial penalty mechanism for companies that cannot meet their environmental commitments may also help prevent them from adopting exaggerated commitments to strengthen their environmental scores from rating providers. Furthermore, the results of the national cultural moderating effect suggest that policymakers should understand the cultural influence on managers' ethical decision-making and adjust climate change-related regulations by considering domestic cultural characteristics. Finally, further analyses emphasize that companies' greenwashing is worsening in societies

with high-power distance characteristics after the Paris Agreement. This is a warning signal, and we suggest that policymakers increase their awareness of this issue and reconsider how to reduce the centralized management style and top management's autocratic power, while adjusting the current regulations. Moreover, encouraging subordinates to express their doubts, especially when corporate practices are inconsistent with ethical principles, may also help prevent MNCs' greenwashing in carbon reduction.

5.4 | Limitations and future research

Regarding limitations and future research, this study focuses only on MNCs listed on the Fortune Global 500 list. However, the climate change crisis requires global collaboration, which means that greenwashing of carbon emissions by small- or medium-sized enterprises (SMEs) is also important. Therefore, investigating SMEs may help policymakers understand whether there are any potential risks and provide suggestions for risk prevention. Moreover, our results reveal that the moderating effects of power distance, long-term orientation, and indulgence are insignificant, indicating that the role of these national cultural dimensions in MNCs' greenwashing is more complex than expected. This implies that a reinvestigation from multiple perspectives is required to extend our knowledge boundaries and provide suggestions for specific cultural contexts. Finally, the moderating effects of individualism between environmental scores and carbon emissions were significant, but not significant for carbon intensity. These findings lead to an interesting question: Why is this different, and are matters of some potential macro-level or micro-level factors causing this phenomenon? This finding warrants further investigation.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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